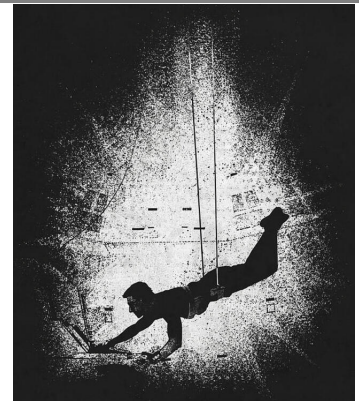




ML Is Possible - The Final Reckoning Part - II



1. Ethan Hunt tasks his IMF member, Luther Stickell, to find a way to linearly separate The Syndicate's non-linear target data (x_i). He proposes using the **Gaussian (RBF) Kernel**, defined as, to map the data into a higher-dimensional feature space $\phi(x)$. We have to prove the theoretical capacity of this weapon by mathematically showing that the feature space $\phi(x)$ implicitly created by the Gaussian Kernel is infinite-dimensional.

To prove $K(x_i, x_j) = \phi(x_i)^T \phi(x_j)$ or $K(x_i, x_j) = \exp\left(-\frac{\|x_i - x_j\|^2}{2\sigma^2}\right)$ maps to infinite dimensions, we analyze the exponential term:

Separate the Interaction Term:

$$K(x_i, x_j) = \exp\left(-\frac{\|x_i\|^2}{2\sigma^2}\right) \exp\left(-\frac{\|x_j\|^2}{2\sigma^2}\right) \underbrace{\exp\left(\frac{x_i^T x_j}{\sigma^2}\right)}_{\text{Interaction term}}$$

Apply Maclaurin Series: We expand the interaction term $\exp(z)$ where $z = \frac{x_i^T x_j}{\sigma^2}$:

$$\begin{aligned} \exp\left(\frac{x_i^T x_j}{\sigma^2}\right) &= \sum_{k=0}^{\infty} \frac{1}{k!} \left(\frac{x_i^T x_j}{\sigma^2}\right)^k \\ &= \sum_{k=0}^{\infty} \left(\frac{1}{\sqrt{k!}\sigma^{2k}} (x_i^T)^k\right) \left(\frac{1}{\sqrt{k!}\sigma^{2k}} (x_j)^k\right) \end{aligned}$$

Because the kernel can be written as an infinite sum of products of terms containing only x_i or only x_j , it is an infinite dot product. This requires the feature vector $\phi(x)$ to have an infinite number of components corresponding to the infinite powers ($k = 0, 1, 2, \dots$) of the original vector x .

2. Benji Dunn is programming a dynamic 'Containment Field' (a Convex Polygon) to trap specific rogue agents (positive class) while leaving civilians (negative class) outside. The field can have an unlimited number of vertices. Determine the capacity of this security system; specifically, what is the **VC Dimension** of the hypothesis class H of all possible Convex Polygons and why? Will the capacity be the same if the hypothesis space is restricted to convex polygons with at most a fixed number of vertices?

The VC Dimension is Infinite. To prove this, we must demonstrate that for any arbitrarily large number of points N , we can find an arrangement of those points that can be "shattered" (perfectly classified for all 2^N possible labeling combinations). Imagine N points positioned on the circumference of a circle. Assign any arbitrary subset of these points as +ive and the remaining points as -ive. Form a polygon by connecting only the +ive points with straight lines. The +ive points are vertices of the polygon (inside). The -ive points are left outside the polygon. This can be done for any labelling and hence the capacity of the model is unbounded.

If you cap the number of vertices, the hypothesis class can no longer shatter arbitrarily large point sets, so its VC dimension becomes finite and depends on that vertex bound k .

3. Luther Stickell is using Bernoulli **Naive Bayes** to classify binary-featured intelligence reports. A single, highly unusual 'Ghost Protocol' message (an outlier) has mistakenly entered the training set for a small class, C_k . Concisely prove how the fundamental probability estimation in Bernoulli NB is immediately vulnerable to this single outlier, demonstrating why the model is not robust.

Bernoulli Naive Bayes is not robust to outliers because its probability estimates are derived directly from simple Maximum Likelihood Estimates (MLE) based purely on raw counts. A single instance can disproportionately shift these ratios, especially when the class size is small.

Probability Definition: The core parameter estimated is the likelihood that a binary feature F_i is present ($F_i = 1$) given the class C_k :

$$P(F_i = 1|C_k) = \frac{(\text{Count of documents in } C_k \text{ with } F_i = 1) + \alpha}{(\text{Total documents in } C_k) + \alpha \cdot 2}$$

(Using Laplace smoothing, α , typically $\alpha = 1$).

Example: Consider a class C_k with only $N = 10$ total documents. Before the outlier, the rare feature F_j has never appeared in C_k .

$$P(\text{No outlier}) : P(F_j = 1|C_k) = \frac{0 + 1}{10 + 2} = \frac{1}{12} \approx 8.3\%$$

Outlier Effect: The outlier message is added, and it possesses the rare feature F_j . The numerator increases by 1, and the total count increases by 1:

$$P(\text{With outlier}) : P(F_j = 1|C_k) = \frac{1 + 1}{10 + 1 + 2} = \frac{2}{13} \approx 15.4\%$$

The single outlier has almost doubled the estimated probability (from 8.3% to 15.4%) that feature F_j belongs to class C_k .

4. The IMF has intercepted two streams of binary data (x_1, x_2) from The Syndicate. A threat is confirmed (Output $y = 1$) only if both streams are active ($x_1 = 1, x_2 = 1$). Otherwise, the output is $y = 0$. Luther Stickell must train a single **Perceptron** to identify this specific threat pattern. Show, step-by-step, the update process for the weights and bias (initially set to zero) until the Perceptron successfully learns the underlying function.

Target function:

$$(x_1 \wedge x_2) = \begin{cases} 0 & \text{if } (x_1, x_2) \in \{(0, 0), (0, 1), (1, 0)\} \\ 1 & \text{if } (x_1, x_2) = (1, 1) \end{cases}$$

Perceptron update rule:

$$\mathbf{w}_{new} = \begin{cases} \mathbf{w}_{old} + \mathbf{x}, & \text{if } y = 0 \text{ and } t = 1 \\ \mathbf{w}_{old} - \mathbf{x}, & \text{if } y = 1 \text{ and } t = 0 \\ \mathbf{w}_{old}, & \text{if } y = t \end{cases}$$

Training pattern order per epoch: (0, 0, 0), (0, 1, 0), (1, 0, 0), (1, 1, 1) where the last value is the target t . Initial weights: $\mathbf{w} = (w_0, w_1, w_2) = (0, 0, 0)$. Bias input: $x_0 = 1$. Input vector: $\mathbf{x} = (1, x_1, x_2)$.

Epoch	(x_1, x_2)	t	$\mathbf{x}$	Before $\mathbf{w}$	net	y	Update	After $\mathbf{w}$
1	(0,0)	0	(1,0,0)	(0,0,0)	0	0	none	(0,0,0)
1	(0,1)	0	(1,0,1)	(0,0,0)	0	0	none	(0,0,0)
1	(1,0)	0	(1,1,0)	(0,0,0)	0	0	none	(0,0,0)
1	(1,1)	1	(1,1,1)	(0,0,0)	0	0	$+\mathbf{x}$	(1,1,1)
2	(0,0)	0	(1,0,0)	(1,1,1)	1	1	$-\mathbf{x}$	(0,1,1)
2	(0,1)	0	(1,0,1)	(0,1,1)	1	1	$-\mathbf{x}$	(-1,1,0)
2	(1,0)	0	(1,1,0)	(-1,1,0)	0	0	none	(-1,1,0)
2	(1,1)	1	(1,1,1)	(-1,1,0)	0	0	$+\mathbf{x}$	(0,2,1)
3	(0,0)	0	(1,0,0)	(0,2,1)	0	0	none	(0,2,1)
3	(0,1)	0	(1,0,1)	(0,2,1)	1	1	$-\mathbf{x}$	(-1,2,0)
3	(1,0)	0	(1,1,0)	(-1,2,0)	1	1	$-\mathbf{x}$	(-2,1,0)
3	(1,1)	1	(1,1,1)	(-2,1,0)	-1	0	$+\mathbf{x}$	(-1,2,1)
4	(0,0)	0	(1,0,0)	(-1,2,1)	-1	0	none	(-1,2,1)
4	(0,1)	0	(1,0,1)	(-1,2,1)	0	0	none	(-1,2,1)
4	(1,0)	0	(1,1,0)	(-1,2,1)	1	1	$-\mathbf{x}$	(-2,1,1)
4	(1,1)	1	(1,1,1)	(-2,1,1)	0	0	$+\mathbf{x}$	(-1,2,2)
5	(0,0)	0	(1,0,0)	(-1,2,2)	-1	0	none	(-1,2,2)
5	(0,1)	0	(1,0,1)	(-1,2,2)	1	1	$-\mathbf{x}$	(-2,2,1)
5	(1,0)	0	(1,1,0)	(-2,2,1)	0	0	none	(-2,2,1)
5	(1,1)	1	(1,1,1)	(-2,2,1)	1	1	none	(-2,2,1)

Final learned weights:

$$\mathbf{w} = (-2, 2, 1)$$

5. Ethan Hunt and the IMF team are attempting to infiltrate a secure facility protected by a 4×4 checkerboard grid defense system. The facility's computer classifies points on this grid based on color: Black = +1 (Access Granted) and White = -1 (Access Denied). The grid spans $0 < x_1 < 4$ (horizontal) and $0 < x_2 < 4$ (vertical). Luther Stickell must design the simplest possible **binary Decision Tree** to perfectly replicate the facility's classification logic. The depth of the tree represents the number of sequential security checks required (minimum number of questions). Note on Boundaries: A point on the border or corner of a region takes the label of the region immediately above it or to its right. For example, (1,0) has label +1, (1,1) has label -1, and (1,1.5) has label -1. Answer the following with justification:

1. What is the minimum depth of the tree that perfectly classifies the colored regions, using only features of the form $x_1 < c$ or of the form $x_2 < c$ for different constants c ?
 2. What is the minimum depth of the tree that perfectly classifies the regions ($0 < x_1 < 4$), using features that only inspect either x_1 or x_2 but not both x_1 and x_2 ?
 3. What is the minimum depth of a binary decision tree that perfectly classifies the regions in checkerboard grid, using ANY features that involve x_1 , x_2 or both?
1. Our 4×4 checkerboard is divided into 16 unit squares, with colors alternating in both directions. Therefore, to classify perfectly, you need at least one leaf per cell i.e., you need at least 16 leaves. A binary decision tree of depth d can have at most 2^d leaves. We need $d \geq 4$.
 2. No single split on a 1-D feature can separate +1 from -1 perfectly for depth 1. The minimum depth that can work (with creative 1-D features) is 2. e.g., $label = (-1)^{colParity + rowParity}$
 3. It can be done with depth 1 since as soon as you allow any feature involving x_1, x_2 , you can hand-engineer a single feature that already encodes the checkerboard pattern e.g., $z(x_1, x_2) = sign(-1^{\lfloor x_1 \rfloor + \lfloor x_2 \rfloor})$